Student Seminar - IDP Presentation 4

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Extraction Grounding

How to make LLM cite it’s extractions?

- Have it provide an evidence span, a text sequence (previous)

- **Have it provide the location on text, where it found the value** (new)

Location Annotations

Sequence/Paragraph/Sentence: We can augment paper full-text with location annotations

[sec-0] Background

[sec-0/p-0]

[sec-0/p-0/s-0] Cell culture techniques are an integral part of cell analysis.

[sec-0/p-0/s-1] Culture methods are often not a primary concern in growth studies, but rather a means to experimental results.

…

[sec-0/p-4]

[sec-0/p-4/s-0] An additional feature of the hydrogel makes it increasingly favorable for use within complex structures.

…

[sec-1] Results and Discussion

[sec-1/sec-0] Hydrogel Formation

[sec-1/sec-0/p-0]

[sec-1/sec-0/p-0/s-0] Hydrogel samples were created from a stock gelling agent of either 25 or 100 mg/ml.

[sec-1/sec-0/p-0/s-1] Successful gelling was performed in purified MilliQ water with both stock solutions.

LLM Grounding

With location annotations, LLM can just point to the location in text:

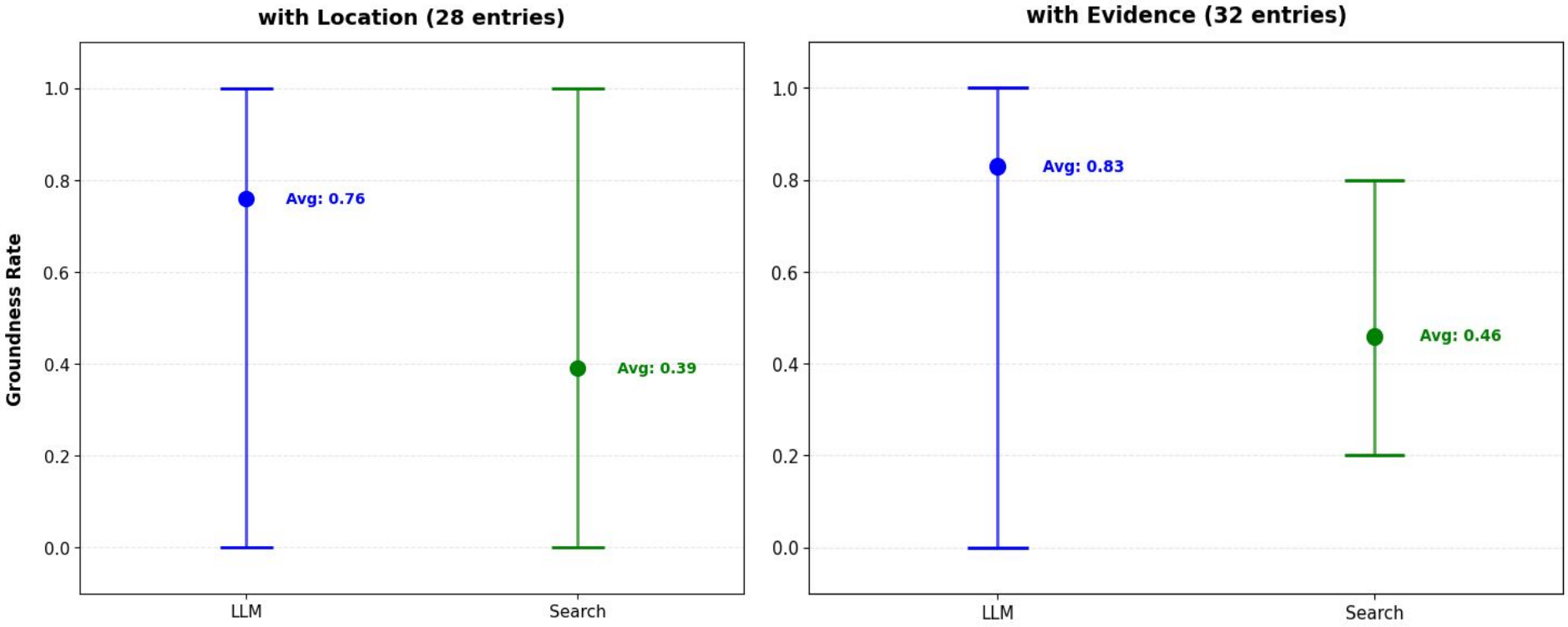
{"sequence":"[sec-1/sec-0/p-0/s-0]",

"type":"[sec-1/sec-1/p-0/s-3]",

"temperature":"[sec-3/sec-15/p-0/s-0]",

"solution":"[sec-1/sec-0/p-0/s-0]"}

Results


with Location (28 entries)
with Evidence (32 entries)
Groundness Rate
1.0
0.8
0.6
0.4
0.2
0.0
Avg: 0.76
Avg: 0.39
Avg: 0.83
Avg: 0.46
LLM
Search

Implementation

Since papers are already in XML, we can directly parse it to object trees:

(RootNode

(SectionNode

(ParagraphNode

(SentenceNode)

(SentenceNode)

…

Structural Nodes:
RootNode
SectionNode
ParagraphNode
SentenceNode

Special Nodes (not implemented yet):
TableNode
ImageNode

Implementation (2)

Benefits of using an object tree:

- Enables location based search
- Provides a single interface over different types of full text content (JATS XML and TEI XML)
- Allows to write utility functions to retrieve parts of text more easily
- Special nodes (table/image) can improve/enable processing of tabular/visual data in the future

Implementation (3)

- For PDF parsing, we can use grobid (https://grobid.readthedocs.io/en/latest/), which returns the pdf in TEI XML format
- For sentence recognition we can use scapy (https://allenai.github.io/scispacy/) with “en_core_sci_sm” model
- Correct sentence segmentation is not essential but should increase extraction performance/grounding (it is done in a best-effort basis)

GPU Access

Another way of running big models: Google Colab (ollama)

- Currently testing, has a limited usage capacity per month but should be useful
- It is meant as a backup for GPU servers / Google AI API

Next Steps

- Identify what's making llm tasks slow and how it can be improved with the current setup (langchain/ollama)
- Run pipeline properly
- Add pdf -> TEI XML -> object tree transformation